

# Graph Neural Network Surrogates for a Coupled Heat-Conduction and Pyrolysis Solver in Thermal Protection System Design

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**Abstract**—Thermal protection system (TPS) design relies on repeated evaluation of an expensive, coupled physics solver that resolves heat conduction, pyrolysis, pyrolysis-gas transport, and aerothermal surface chemistry. We present early work toward a graph neural network (GNN) surrogate that learns the per-step evolution of a 1D heat-shield pyrolysis solver. As a foundation, we re-implement a trusted Fortran reference solver in modular, vectorized Python and verify it against the original by column-level comparison of its thermocouple and density outputs, matching to machine precision on a conduction-pyrolysis case and to 0.064 K on a full gas and aerothermal case. Training on data from the validated solver, a per-cell message-passing GNN reproduces a held-out heating scenario over a full 60 s rollout to a mean error of 7.6 K and tracks the pyrolysis front to within a fraction of a millimeter.

**Index Terms**—graph neural networks, surrogate models, thermal protection systems, pyrolysis, scientific machine learning

## I. INTRODUCTION

Thermal protection systems shield a vehicle from the intense aerothermal heating of atmospheric entry. *[draft body from white paper; add citations at marked spots.]* Our surrogate follows the mesh-based graph-network line of work [1], [2], applied here to a multi-physics heat-shield solver.

## II. REFERENCE SOLVER

$$\rho c_p \frac{\partial T}{\partial t} = \frac{\partial}{\partial x} \left( k \frac{\partial T}{\partial x} \right) + (\text{pyrolysis and gas source terms}). \quad (1)$$

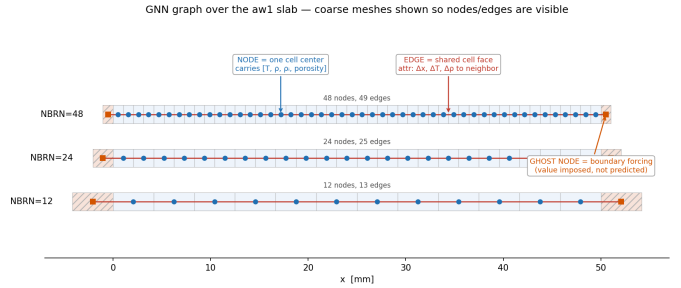


Fig. 1. The mesh as a graph: each finite-volume cell is a node carrying  $[T, \rho, p, \text{porosity}]$ ; edges join neighboring cells with relative features (neighbor-minus-self state and the cell spacing  $\Delta x$ ). Ghost nodes at the ends impose the boundary forcing. Weights are shared across all cells. (Coarse meshes shown for legibility.)

TABLE I  
PORT VALIDATION AGAINST THE FORTRAN REFERENCE (AW2\_TC21).

| Quantity                | Agreement                                   |
|-------------------------|---------------------------------------------|
| All temperature columns | max 0.064 K, mean $\sim 4 \times 10^{-4}$ K |
| Hot aerothermal surface | 0.06 K                                      |
| Densities               | $\sim 10^{-3}$ to $10^{-9}$                 |
| Time, cold back face    | bit-exact                                   |

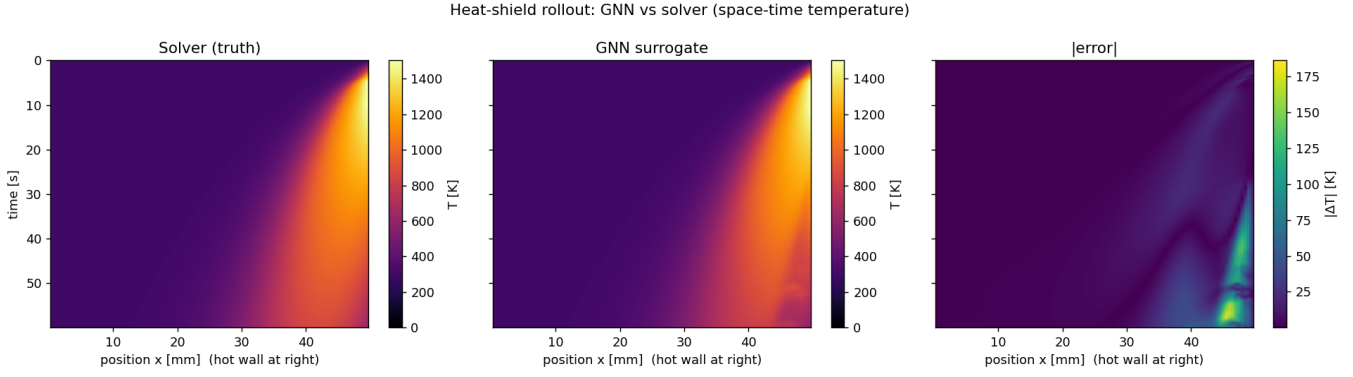


Fig. 2. aw1 surrogate rollout (gas off), space-time temperature: solver truth (left), GNN surrogate (center), absolute error (right). Mean rollout error 7.6 K over the full 60 s; the surrogate field is visually indistinguishable from the solver and the error is small everywhere.

Why aw1 works and aw2 fails: a fast surface transient the coarse super-step can't track (temporal, not spatial)

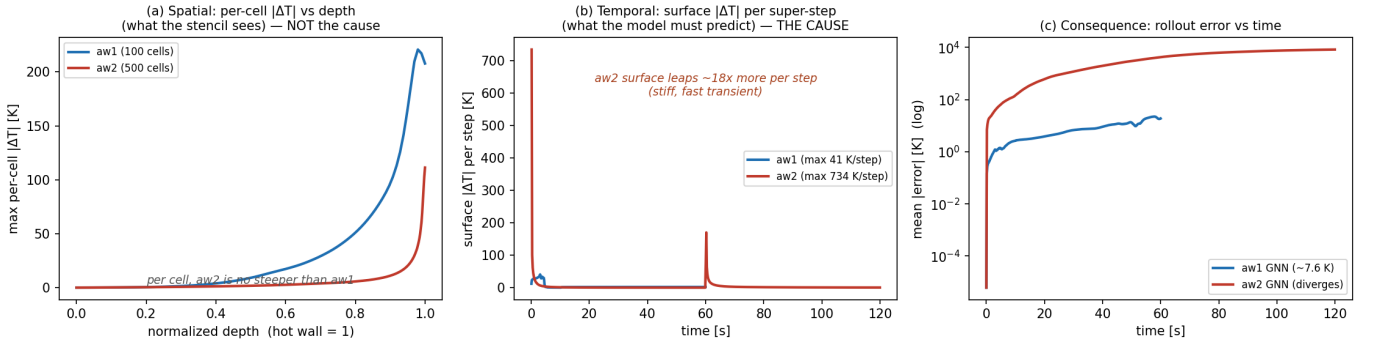


Fig. 3. Why aw2 fails and aw1 does not, and why the cause is temporal not spatial. (a) Per-cell spatial  $|\Delta T|$ : aw2 is no steeper than aw1. (b) Per-super-step surface  $|\Delta T|$ : aw2 leaps  $\sim 18\times$  more (734 vs 41 K) under the square-pulse aerothermal flux. (c) The consequence: aw1 rollout error stays  $\sim 7.6$  K while aw2 diverges.

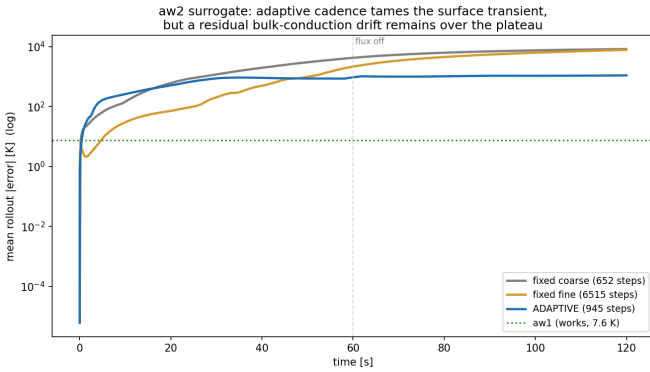


Fig. 4. aw2 rollout error vs time across recipes. Fixed coarse and fixed fine cadence both diverge; the  $\Delta t$ -aware adaptive schedule stays bounded. All shown in-sample; held-out, even the adaptive model does not yet generalize (Section IV-C).

### III. METHODOLOGY

- A. Python Port
- B. Verification
- C. Training-Data Generation
- D. GNN Architecture

### IV. RESULTS

- A. Port Validation
- B. Surrogate Accuracy (aw1)
- C. Gas Case (aw2): A Diagnosed Failure Mode

program within the group.

### REFERENCES

- [1] T. Pfaff, M. Fortunato, A. Sanchez-Gonzalez, and P. W. Battaglia, "Learning mesh-based simulation with graph networks," in *International Conference on Learning Representations (ICLR)*, 2021.
- [2] A. Sanchez-Gonzalez, J. Godwin, T. Pfaff, R. Ying, J. Leskovec, and P. W. Battaglia, "Learning to simulate complex physics with graph networks," in *International Conference on Machine Learning (ICML)*, 2020.